

PilotApply

How I Built an AI-Powered Job Application Assistant

pilotapply.web.app

Live job search across 10 platforms

AI-generated cover letters with Gemini

Auto-filled application form answers

Multi-profile system with cloud sync

Google Authentication + Firebase backend

React Firebase Gemini AI Adzuna API

A complete guide — from zero to deployed

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Introduction

Job hunting is exhausting. Every application requires a tailored cover letter, repeated form filling, and hours spent searching across multiple job boards. I built PilotApply to solve exactly this problem — an AI-powered assistant that handles the repetitive parts of job applications so you can focus on what matters.

This guide explains exactly how I built it — every tool, every API, every decision. If you want to build something similar or contribute to this project, this is your complete reference.

What PilotApply Does

- Searches live jobs across 10 platforms simultaneously
- Generates personalized cover letters using Gemini AI in seconds
- Auto-fills 7 common application form questions tailored to the job
- Manages multiple profiles for different roles (e.g. Cloud Engineer, Full Stack Developer)
- Saves everything to the cloud — works on any device after Google sign-in

Tech Stack at a Glance

Layer	Technology	Why I chose it	Cost
Frontend	React.js	Component-based, fast, huge ecosystem	Free
Hosting	Firebase Hosting	One command deploy, global CDN	Free
Database	Firebase Firestore	Real-time sync, no backend needed	Free tier
Auth	Firebase Auth	Google login in 10 lines of code	Free
Backend	Firebase Cloud Functions	Secure API proxy without a server	Free tier
Job Search	Adzuna API	500 free requests/month, no card needed	Free
AI	Google Gemini AI	1500 free requests/day, powerful model	Free tier

TIP: The entire app runs for free within these service limits for personal use. You would need thousands of daily users before incurring any charges.

Chapter 1: Prerequisites & Setup

Before writing any code, you need to set up accounts and install tools. Here is everything you need:

1.1 Accounts to Create

Google Account	Used for Firebase, Gemini AI, and Google login in the app
Firebase Project	console.firebase.google.com — create a new project
Adzuna Account	developer.adzuna.com — free job search API, no card needed
GitHub Account	github.com — to push your code publicly

1.2 Tools to Install

Step 1: Install Node.js

- Go to nodejs.org and download the LTS version
- Verify installation:

```
node --version
```

```
npm --version
```

Step 2: Install Firebase CLI

- Run this command in your terminal:

```
sudo npm install -g firebase-tools
```

- Verify it installed:

```
firebase --version
```

Step 3: Create a React App

- Navigate to where you want your project:

```
npx create-react-app job-assistant
```

```
cd job-assistant
```

- Delete the default files you don't need:

```
rm src/App.js src/App.test.js src/logo.svg
```

Chapter 2: Setting Up Firebase

Firebase is the backbone of this app — it handles authentication, database, hosting, and our secure backend functions. Here is how to set it up from scratch.

2.1 Create Your Firebase Project

Step 1: Create the project

- Go to console.firebase.google.com
- Click 'Add project' and name it (e.g. job-assistant)
- Disable Google Analytics if you want — not needed for this app
- Click 'Create project'

Step 2: Add a Web App

- Inside your project, click the (Web) icon
- Name it (e.g. job-assistant-web)
- Check 'Also set up Firebase Hosting'
- Click 'Register app'
- Copy the `firebaseConfig` object — you'll need this later

Step 3: Enable Firestore Database

- Left sidebar → Firestore Database → Create database
- Choose 'Start in test mode'
- Select region: asia-south1 (closest to India)
- Click Enable and wait ~30 seconds

Step 4: Enable Google Authentication

- Left sidebar → Authentication → Get started
- Click 'Google' from the provider list
- Toggle Enable to ON
- Select your support email from the dropdown
- Click Save

Step 5: Upgrade to Blaze Plan (for Cloud Functions)

- Left sidebar → Usage and billing → Modify plan
- Select Blaze (pay-as-you-go) — required for Cloud Functions
- Add your card — you will NOT be charged within free limits
- Set a budget alert of \$100 for safety

WARNING: Cloud Functions require the Blaze plan. However the free tier includes 2 million function calls per month — you will not be charged for normal personal use.

2.2 Install Firebase in Your Project

```
npm install firebase
```

Create src/firebase.js with your project config:

```
import { initializeApp } from 'firebase/app';
import { getFirestore } from 'firebase/firestore';
import { getAuth, GoogleAuthProvider } from 'firebase/auth';

const firebaseConfig = {
  apiKey: process.env.REACT_APP_FIREBASE_API_KEY,
  authDomain: process.env.REACT_APP_FIREBASE_AUTH_DOMAIN,
  projectId: process.env.REACT_APP_FIREBASE_PROJECT_ID,
  storageBucket: process.env.REACT_APP_FIREBASE_STORAGE_BUCKET,
  messagingSenderId: process.env.REACT_APP_FIREBASE_MESSAGING_SENDER_ID,
  appId: process.env.REACT_APP_FIREBASE_APP_ID,
};

const app = initializeApp(firebaseConfig);
export const db = getFirestore(app);
export const auth = getAuth(app);
export const provider = new GoogleAuthProvider();
```

TIP: Notice we use process.env.* instead of hardcoded values. This keeps your keys out of GitHub. Store actual values in a .env.local file that you never push.

Chapter 3: Getting Your API Keys

The app needs three sets of credentials. Here is exactly where to get each one and how to use them safely.

3.1 Adzuna API — Job Search

Website	developer.adzuna.com
Free limit	500 requests per month
Card needed	No — completely free to sign up
What you get	Application ID and Application Key

Step 1: Get Your Adzuna Keys

- Go to developer.adzuna.com and click Register Now
- Fill in your details — no credit card required
- Once logged in, look for 'Create App' or 'Dashboard'
- Name your app (e.g. pilotapply) and create it
- Copy your Application ID and Application Key
- Paste them into your functions/.env file (see Chapter 4)

3.2 Gemini AI API — Cover Letters & Autofill

Website	aistudio.google.com
Free limit	1,500 requests per day
Card needed	No — if created in a fresh project
What you get	API Key

Step 2: Get Your Gemini API Key

- Go to aistudio.google.com
- Click 'Get API key' in the left sidebar
- Click 'Create API key'
- IMPORTANT: Select 'Create API key in new project'
- Copy the generated key
- Paste it into your functions/.env file

WARNING: Do NOT create the Gemini key inside your Firebase project. It causes quota issues with limit: 0 errors. Always create it in a separate new project at aistudio.google.com.

3.3 Firestore Security Rules

After enabling Firestore, replace the default rules to ensure each user can only access their own data:

```
rules_version = '2';
service cloud.firestore {
  match /databases/{database}/documents {
    match /users/{userId}/{document=**} {
      allow read, write: if request.auth != null
      && request.auth.uid == userId;
    }
  }
}
```

Apply these in Firebase Console → Firestore → Rules tab → Publish.

Chapter 4: Cloud Functions — The Secure Backend

This is one of the most important parts of the architecture. Direct API calls from the browser are blocked by CORS, and hardcoded API keys are visible to anyone who inspects the page. Cloud Functions solve both problems.

4.1 Why Cloud Functions?

Problem	Without Functions	With Functions
CORS errors	Browser blocks direct API calls	Server-side calls work fine
API key security	Keys visible in browser source code	Keys hidden on Google's servers
Reliability	CORS proxies go down randomly	Google infrastructure — 99.9% uptime

4.2 Setting Up Cloud Functions

Step 1: Initialize Functions

```
firebase init functions
```

- Select JavaScript when asked for language
- Type N for ESLint
- Type Y to install dependencies
- Wait for 'Firebase initialization complete'

Step 2: Create functions/.env with your keys

- Create a file called .env inside the functions/ folder:

```
ADZUNA_APP_ID=your_id_here
```

```
ADZUNA_APP_KEY=your_key_here
```

```
GEMINI_API_KEY=your_key_here
```

- This file is never pushed to GitHub — add it to .gitignore

4.3 The Cloud Function Code

Replace everything in `functions/index.js` with two functions — one for job search and one for AI generation:

```
const functions = require('firebase-functions');
const https = require('https');

const ADZUNA_APP_ID = process.env.ADZUNA_APP_ID;
const ADZUNA_APP_KEY = process.env.ADZUNA_APP_KEY;
const GEMINI_API_KEY = process.env.GEMINI_API_KEY;
const GEMINI_MODEL = 'gemini-2.0-flash';

// searchJobs — proxies Adzuna API calls
exports.searchJobs = functions.https.onRequest((req, res) => {
  res.set('Access-Control-Allow-Origin', '*');
  // ... builds URL with your keys and fetches from Adzuna
});

// generateAI — proxies Gemini API calls
exports.generateAI = functions.https.onRequest(async (req, res) => {
  res.set('Access-Control-Allow-Origin', '*');
  // ... sends prompt to Gemini and returns response
});
```

Step 3: Deploy the Functions

```
firebase deploy --only functions
```

- Wait 2-3 minutes for deployment
- Copy the two Function URLs from the output
- These URLs go into your React hooks (`useJobSearch.js` and `useAIGeneration.js`)
- If you see v2 functions, get the correct URL from Firebase Console → Functions

TIP: If you only update `functions/index.js`, you don't need to rebuild React. Just run: `firebase deploy --only functions`

Chapter 5: Project Architecture

The app is split into small, focused files. Here is what each one does and why it exists:

5.1 File Structure

File	Purpose
src/firebase.js	Initializes Firebase — imported once, used everywhere
src/App.jsx	Root component — handles auth state and tab routing
src/components/LoginScreen.jsx	Google sign-in screen shown to logged-out users
src/components/Header.jsx	Navigation tabs + user avatar + sign out
src/components/ProfileTab.jsx	Multi-profile management UI
src/components/FindJobsTab.jsx	Job search UI with platform toggles
src/components/JobCard.jsx	Individual job listing card
src/components/CoverLetterTab.jsx	Cover letter generation UI
src/components/ApplicationKitTab.jsx	Autofill answers generation UI
src/hooks/useProfiles.js	All Firestore CRUD for profiles
src/hooks/useJobSearch.js	Calls searchJobs Cloud Function
src/hooks/useAIGeneration.js	Calls generateAI Cloud Function
src/constants/index.js	Platform list, date filters, default profile
src/utils/helpers.js	matchScore, timeAgo, copyToClipboard
src/styles/global.css	All CSS styles
functions/index.js	Cloud Functions — Adzuna + Gemini proxy

5.2 Data Flow

Job Search	User types query → React calls Cloud Function URL → Function calls Adzuna API → Results returned to UI
Cover Letter	User clicks Generate → React sends job + profile to Cloud Function → Function calls Gemini → Letter returned

Profile Save	User types in profile → React writes to Firestore in real time → Synced across all devices
Authentication	User clicks Google → Firebase Auth popup → Google verifies → User ID used for Firestore paths

Chapter 6: Deploying the App

Deployment takes three commands and about two minutes. Here is the complete process:

6.1 First Time Setup

Step 1: Login to Firebase

```
firebase login
```

- A browser window opens — sign in with your Google account
- Wait for 'Success! Logged in as your@email.com' in terminal

Step 2: Initialize Firebase in Your Project

```
firebase init
```

- Select Firestore and Hosting using spacebar
- Choose 'Use an existing project' → select your project
- When asked for public directory, type: build
- Configure as single-page app: Y
- Set up automatic builds with GitHub: N
- Wait for 'Firebase initialization complete'

Step 3: Set Custom Site Name (Optional)

- Firebase Console → Hosting → Add another site
- Type your preferred name (e.g. pilotapply)
- Back in terminal: `firebase target:apply hosting pilotapply pilotapply`
- Update `firebase.json` to include: `"target": "pilotapply"`

6.2 Every Time You Redeploy

Command	When to use
<code>npm start</code>	Test locally before deploying
<code>npm run build</code>	Build production version — always run before deploying
<code>firebase deploy</code>	Deploy everything — hosting + functions
<code>firebase deploy --only hosting</code>	Only deploy React app changes

```
firebase deploy --only functions
```

Only deploy Cloud Function changes

WARNING: Always run `npm run build` before `firebase deploy`. If you skip this step, the old version of your app gets deployed.

6.3 Add Your Domain to Auth

After deploying with a custom domain, add it to Firebase Auth authorized domains:

Step 4: Authorize Your Domain

- Firebase Console → Authentication → Settings tab
- Click Authorized domains → Add domain
- Type your domain (e.g. `pilotapply.web.app`)
- Click Add — sign-in will now work on your live URL

Chapter 7: Taking Down the Application

There are different levels of 'taking down' depending on what you want. Here are all your options:

7.1 Temporarily Disable

The app goes offline but all your data is kept safe. You can bring it back anytime.

```
firebase hosting:disable
```

Or from Firebase Console → Hosting → click your site → Disable.

To bring it back:

```
npm run build && firebase deploy
```

7.2 Delete the Site

Removes the URL permanently. Your Firestore data and Cloud Functions are still running.

What gets deleted	The website at pilotapply.web.app
What stays	Firestore data, Auth users, Cloud Functions
URL reusable?	No — once deleted, the subdomain is gone permanently
How to redo	Create a new site name and redeploy

7.3 Stop Billing Completely

Step 1: Downgrade from Blaze to Spark

- Firebase Console → Usage and billing → Modify plan
- Select Spark (free) → Confirm
- Note: Cloud Functions will stop working on Spark plan

Step 2: Remove Your Card

- Go to console.cloud.google.com/billing
- Click your billing account → Account management
- Click 'Cancel billing account'
- Any unused credits are refunded

7.4 Delete Everything

WARNING: This is permanent and irreversible. All user data, profiles, and the app will be completely gone.

Step 3: Delete the Firebase Project

- Firebase Console → your project → Settings (gear icon)
- Scroll to bottom → 'Delete project'
- Type the project name to confirm
- Everything is deleted — Firestore, Auth, Hosting, Functions

Chapter 8: Pushing to GitHub Safely

Before pushing your code to a public GitHub repository, you must remove all API keys and secrets. Here is the complete checklist:

8.1 Files to Never Push

Create a .gitignore file in your project root with these entries:

```
node_modules/
build/
.env
.env.local
.env.production
functions/.env
.firebase/
```

8.2 Files That Need Changes Before Pushing

File	Remove	Replace with
src/firebase.js	Actual firebaseConfig values	process.env.REACT_APP_* variables
functions/index.js	Actual API keys	process.env.* variables
src/hooks/useJobSearch.js	Nothing — Cloud Function URL is safe	No change needed
src/hooks/useAIGeneration.js	Nothing — Cloud Function URL is safe	No change needed

8.3 Environment Variables Setup

Step 1: Create .env.local in project root

- REACT_APP_FIREBASE_API_KEY=your_actual_key
- REACT_APP_FIREBASE_AUTH_DOMAIN=your_project.firebaseio.com
- REACT_APP_FIREBASE_PROJECT_ID=your_project_id
- REACT_APP_FIREBASE_STORAGE_BUCKET=your_project.appspot.com
- REACT_APP_FIREBASE_MESSAGING_SENDER_ID=your_sender_id
- REACT_APP_FIREBASE_APP_ID=your_app_id
- This file stays on your Mac — it is never pushed to GitHub

Step 2: Create functions/.env

- ADZUNA_APP_ID=your_adzuna_id
- ADZUNA_APP_KEY=your_adzuna_key
- GEMINI_API_KEY=your_gemini_key
- This file also stays local — never pushed

Step 3: Verify before pushing

```
cat src/firebase.js
```

- Should show process.env.* — not actual keys

```
cat functions/index.js
```

- Should show process.env.* — not actual keys

```
cat .gitignore
```

- Should include .env.local and functions/.env

TIP: The Cloud Function URLs (like <https://searchjobs-xxx.run.app>) are completely safe to push publicly. They are just URLs — the actual API keys are hidden inside Google's servers.

Chapter 9: Cost Breakdown

Here is an honest breakdown of what this app costs to run and where charges could theoretically appear:

Service	Free Limit	Typical Usage	Charge Risk
Firebase Hosting	10GB storage, 360MB/day transfer	Less than 1MB/day	None
Firebase Firestore	50,000 reads, 20,000 writes per day	About 50 reads/day	None
Firebase Auth	Unlimited users	Personal use only	None
Firebase Functions	2 million calls per month	About 200 per month	None
Adzuna API	500 requests per month	About 100 per month	None
Gemini AI API	1,500 requests per day	About 10 per day	None

TIP: For personal use or sharing with a few friends, this app is completely free. You would need thousands of daily active users before any charges appear.

Budget Alert

Set a budget alert so you always know before any charges occur:

Step 1: Set Budget Alert in Firebase

- Firebase Console → your project → Usage and billing
- Click 'Set budget alert'
- Set amount to ■100
- You will receive an email before any charge ever happens

Chapter 10: Quick Reference

Important URLs

Live App	https://pilotapply.web.app
Firebase Console	https://console.firebase.google.com
Google Cloud Console	https://console.cloud.google.com
Adzuna Dashboard	https://developer.adzuna.com
Gemini AI Studio	https://aistudio.google.com
Cloud Function — Jobs	https://searchjobs-sg6cmaia7q-uc.a.run.app
Cloud Function — AI	https://generateai-sg6cmaia7q-uc.a.run.app

Most Used Commands

Command	What it does
npm start	Run app locally at localhost:3000
HOST=0.0.0.0 npm start	Run locally — accessible on phone via same WiFi
npm run build	Build production version
firebase deploy	Deploy everything
firebase deploy --only functions	Deploy only backend changes
firebase deploy --only hosting	Deploy only frontend changes
firebase functions:list	List deployed functions and their URLs
firebase functions:log	View Cloud Function logs for debugging
firebase hosting:disable	Take the app offline temporarily

Troubleshooting

Error	Most Likely Cause	Fix
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Failed to fetch jobs	CORS or function URL wrong	Check firebase functions:list for correct URL
No response (cover letter)	Gemini key or wrong model name	Check functions/index.js keys and GEMINI_MODEL value
Sign-in failed	Domain not authorized	Firebase Auth → Settings → Add domain
Module not found	File missing or wrong path	Check file exists in correct folder
HTTP 400	Wrong parameters sent	Check params in useJobSearch.js
HTTP 404	Wrong Cloud Function URL	Run firebase functions:list
limit: 0 (Gemini)	Key created in wrong project	Create new key at aistudio.google.com in new project
EACCES permissions error	npm permissions on Mac	Add sudo before the command

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